expense of connectivity: 20,000 years ago there were many more Sapiens than 70,000 years ago, and Sapiens in Europe processed information differently to Sapiens in China. However, there were no connections between people in Europe and China, and it would have seemed utterly impossible that all Sapiens may one day be part of a single data-processing web.

The second stage began with the Agricultural Revolution and continued until the invention of writing and money about 5,000 years ago. Agriculture speeded demographic growth, so the number of human processors rose sharply. Simultaneously, agriculture enabled many more people to live together in the same place, thereby generating dense local networks that contained an unprecedented number of processors. In addition, agriculture created new incentives and opportunities for different networks to trade and communicate with one another. Nevertheless, during the second phase centrifugal forces remained predominant. In the absence of writing and money, humans could not establish cities, kingdoms or empires. Humankind was still divided into innumerable little tribes, each with its own lifestyle and world view. Uniting the whole of humankind was not even a fantasy.

The third stage kicked off with the invention of writing and money about 5,000 years ago, and lasted until the beginning of the Scientific Revolution. Thanks to writing and money, the gravitational field of human cooperation finally overpowered the centrifugal forces. Human groups bonded and merged to form cities and kingdoms. Political and commercial links between different cities and kingdoms also tightened. At least since the first millennium BC – when coinage, empires and universal religions appeared – humans began to consciously dream about forging a single network that would encompass the entire globe.

This dream became a reality during the fourth and last stage of history, which began around 1492. Early modern explorers, conquerors and traders wove the first thin threads that encompassed the whole world. In the late modern period these threads were made stronger and denser, so that the spider's web of Columbus's days became the steel and asphalt grid of the twenty-first century. Even more importantly, information was allowed to flow increasingly freely along this global grid. When